

Exploratory Data Analysis on IMDb

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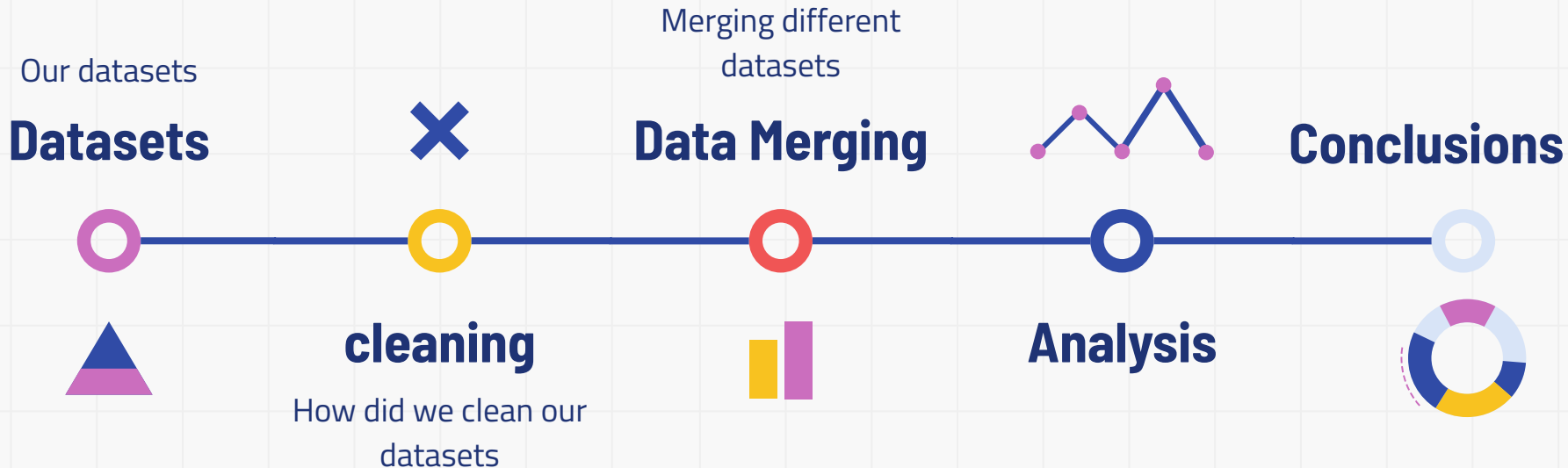


Android Tablets

IMDb

Find Showtimes | Watch Trailers | Rate Movies

Statistical process



Our Datasets



title.basics

Main title information:

tconst: Title ID

primaryTitle, originalTitle: Popular & original names

titleType: movie, tvSeries, etc.

startYear, runtimeMinutes, genres



title.ratings

User ratings:

tconst

averageRating, numVotes



title.akas

Alternate/localized titles:

titleId: Link to tconst

title: Localized title

region, language

types, attributes, isOriginalTitle



title.crew

Creators:

tconst: Title ID

directors, writers: Person IDs (nconst)



title.episode

TV Episode mapping:

tconst: Episode ID

parentTconst: Series ID

seasonNumber, episodeNumber



title.principals

Cast & crew roles:

tconst, nconst: Title & person ID

category: actor, director, etc.

job, characters



name.basics

Information about people:

nconst: Person ID

primaryName, birthYear, deathYear

primaryProfession, knownForTitles: Title IDs





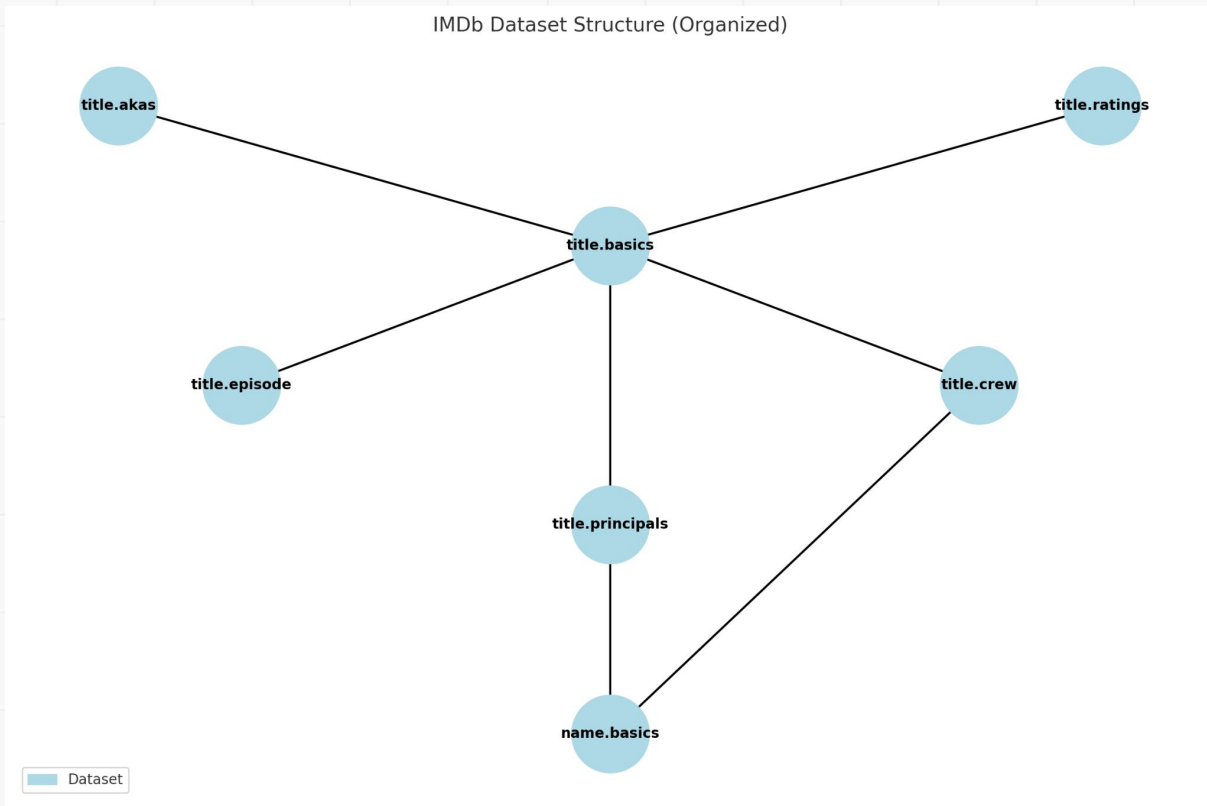
11,000,000+

Number of Rows



Datasets Understanding

IMDb Dataset Structure (Organized)



Data Cleaning

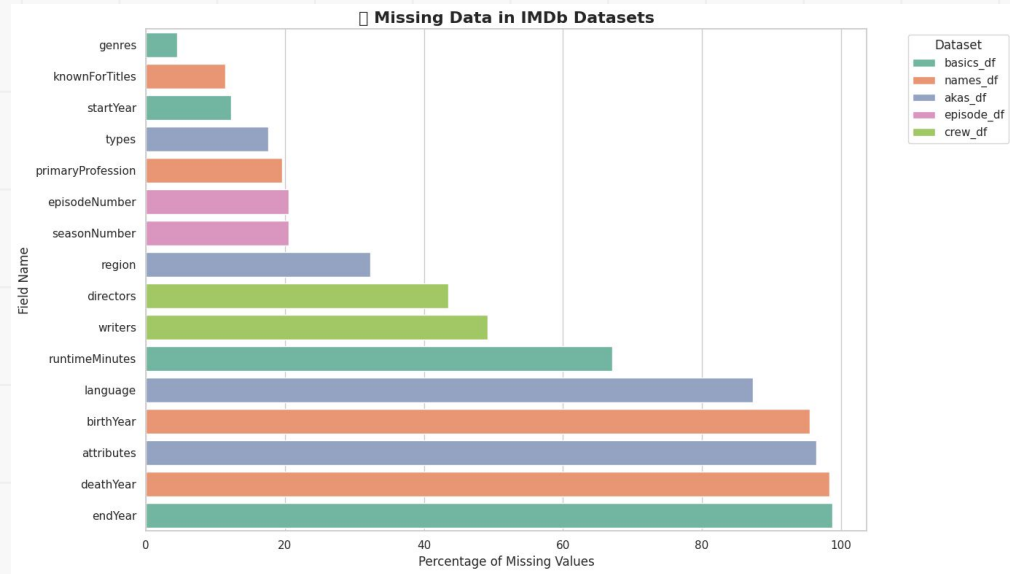
Handled missing values ($\backslash N \rightarrow \text{NaN}$)

Removed irrelevant types (e.g., tvEpisode)

Split lists (genres, professions) for analysis

Dropped duplicate and low-information columns

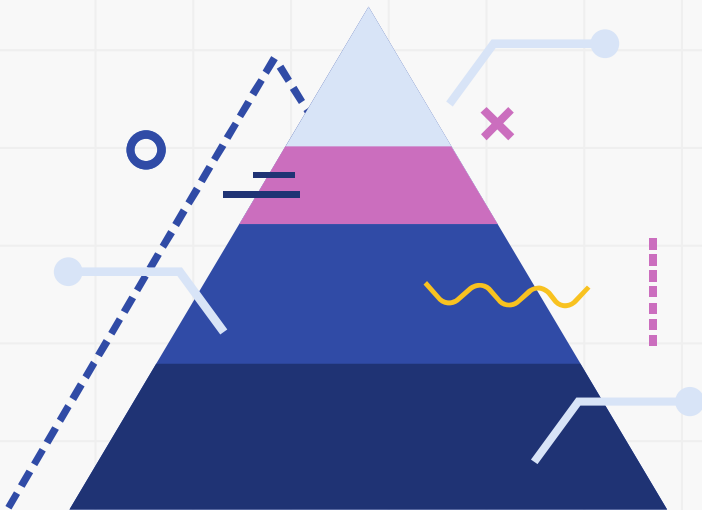
Ensured consistent formats (e.g., numeric years)



Data Merging

No full merge due to memory size

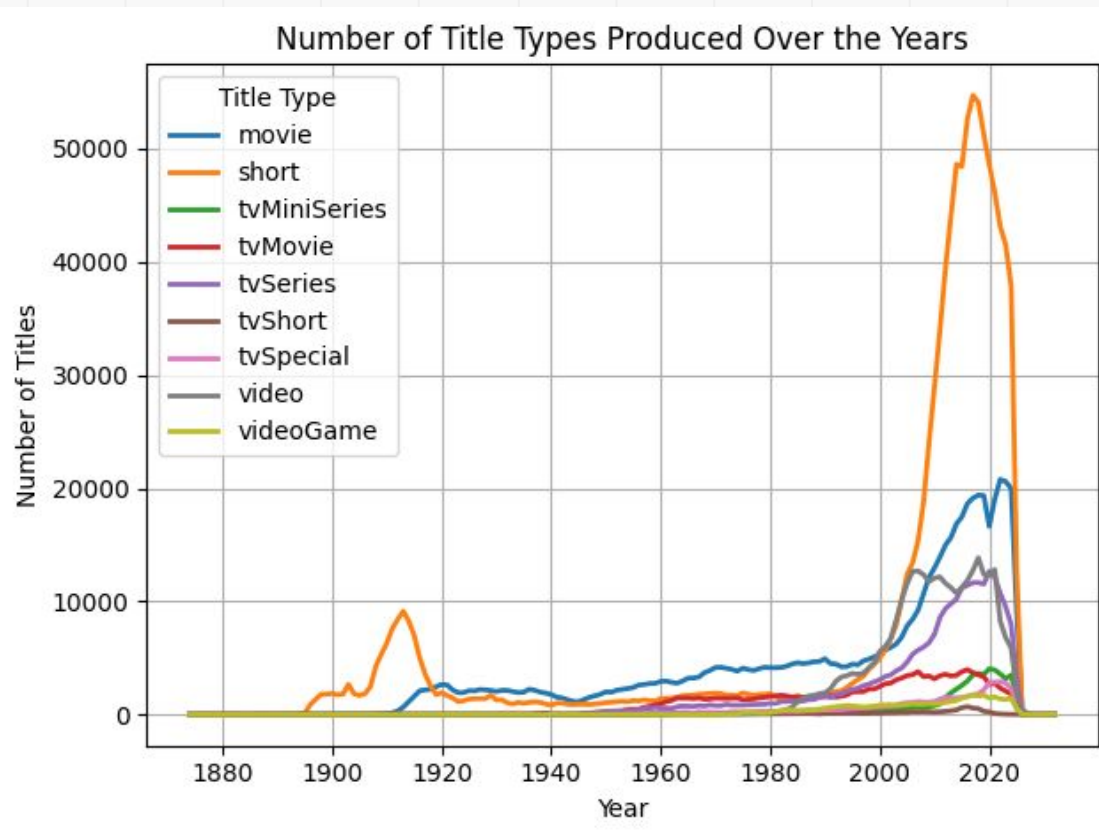
Created goal-specific Data Frames for each research question

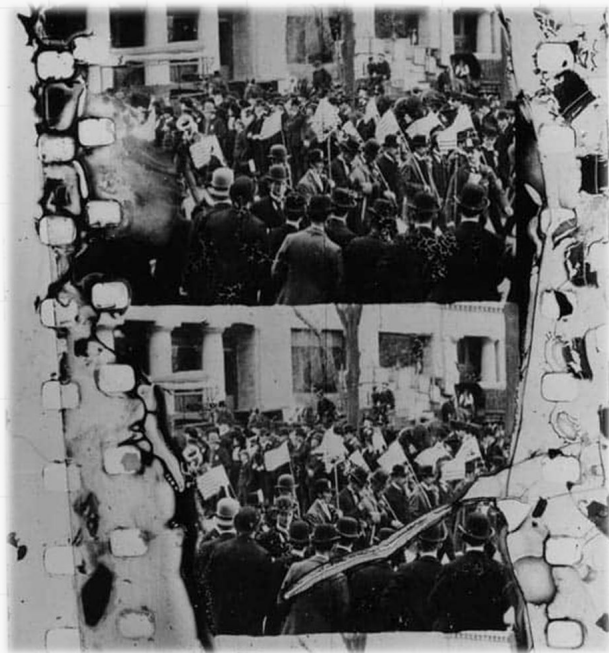
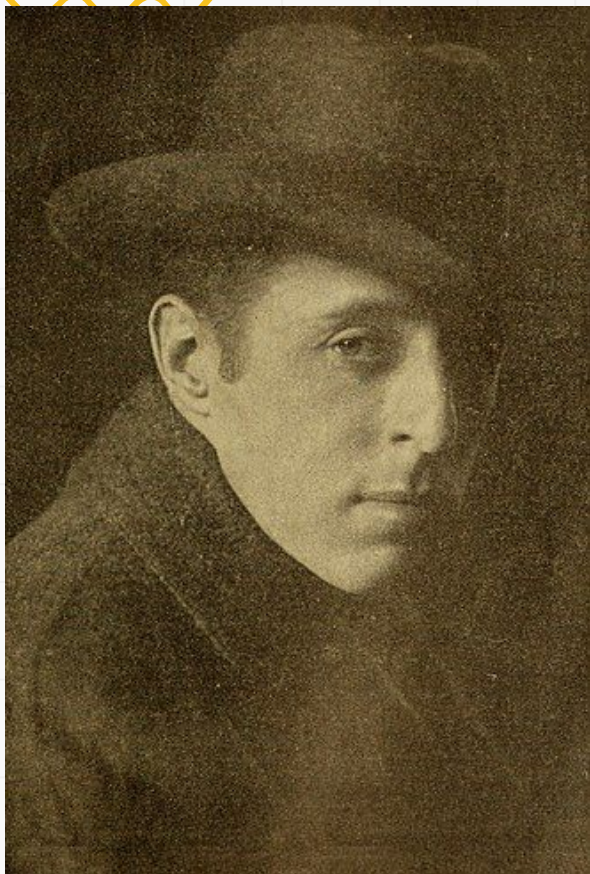


Dataset Name	Purpose	Source Tables
df_title_trends	Title type counts & trends	title.basics
df_ratings_analysis	Average rating by type, year, genre	title.basics + title.ratings
df_votes_analysis	Votes distribution, rating correlation	title.basics + title.ratings
df_genre_counts	Genre counts, combinations	title.basics
df_genre_ratings	Ratings by genre	title.basics + title.ratings
df_directors	Frequent and best-rated directors	title.crew + name.basics (+ ratings)
df_writers	Frequent and best-rated writers	title.crew + name.basics (+ ratings)
df_actors	Frequent and best-rated actors	title.principals + name.basics (+ ratings)
df_akas_regions	Regional title localization patterns	title.akas

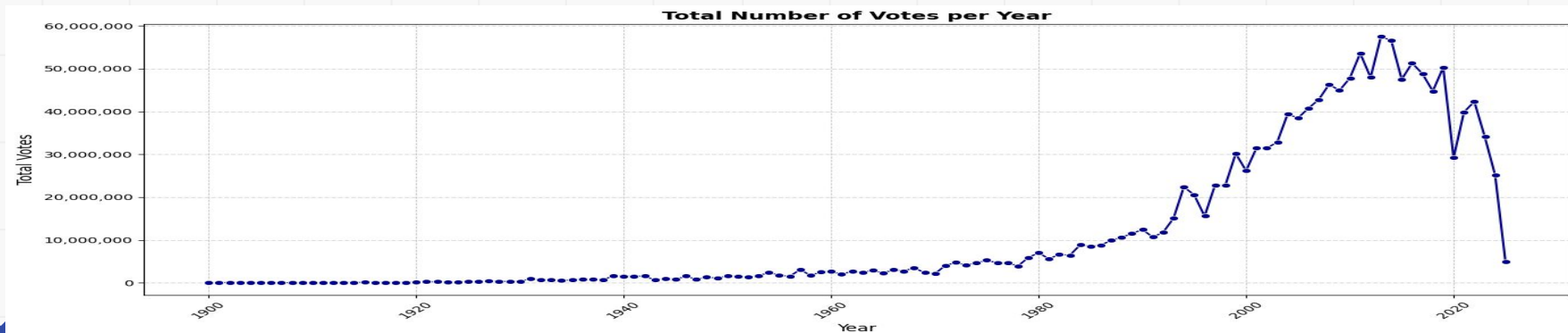


Title through the years

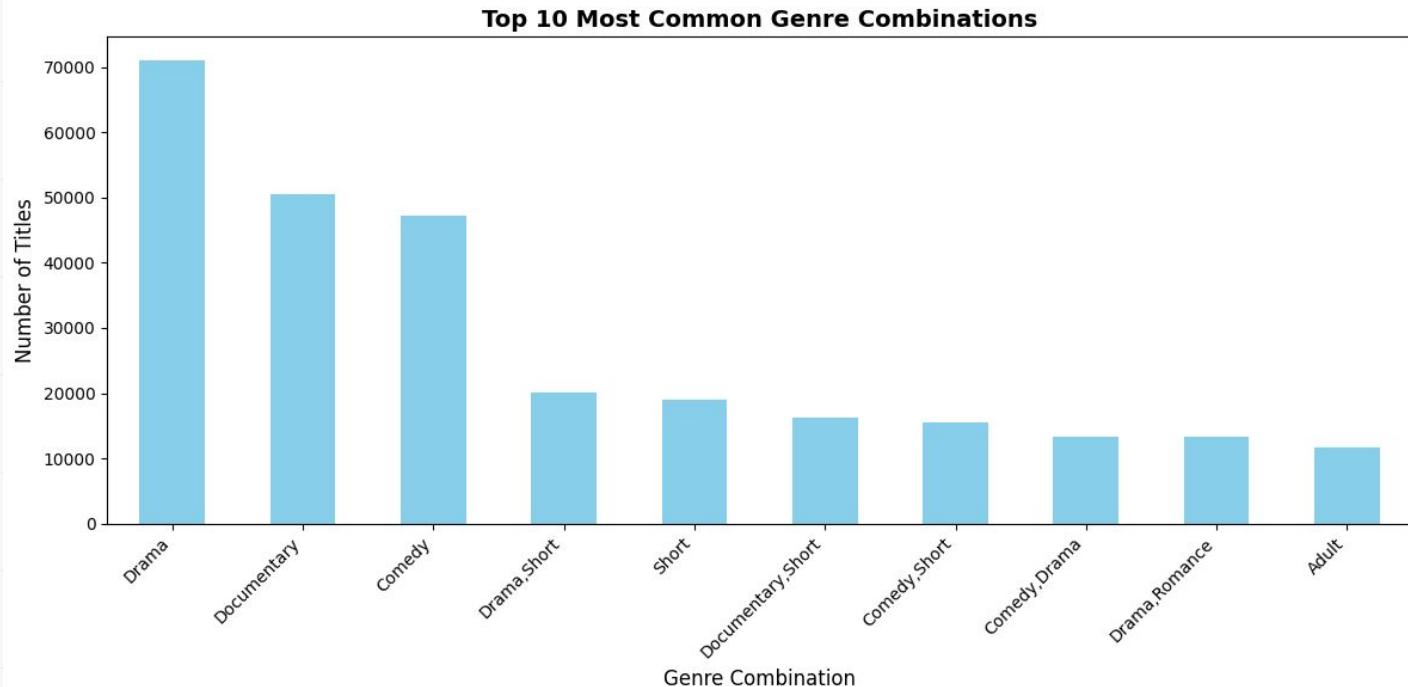




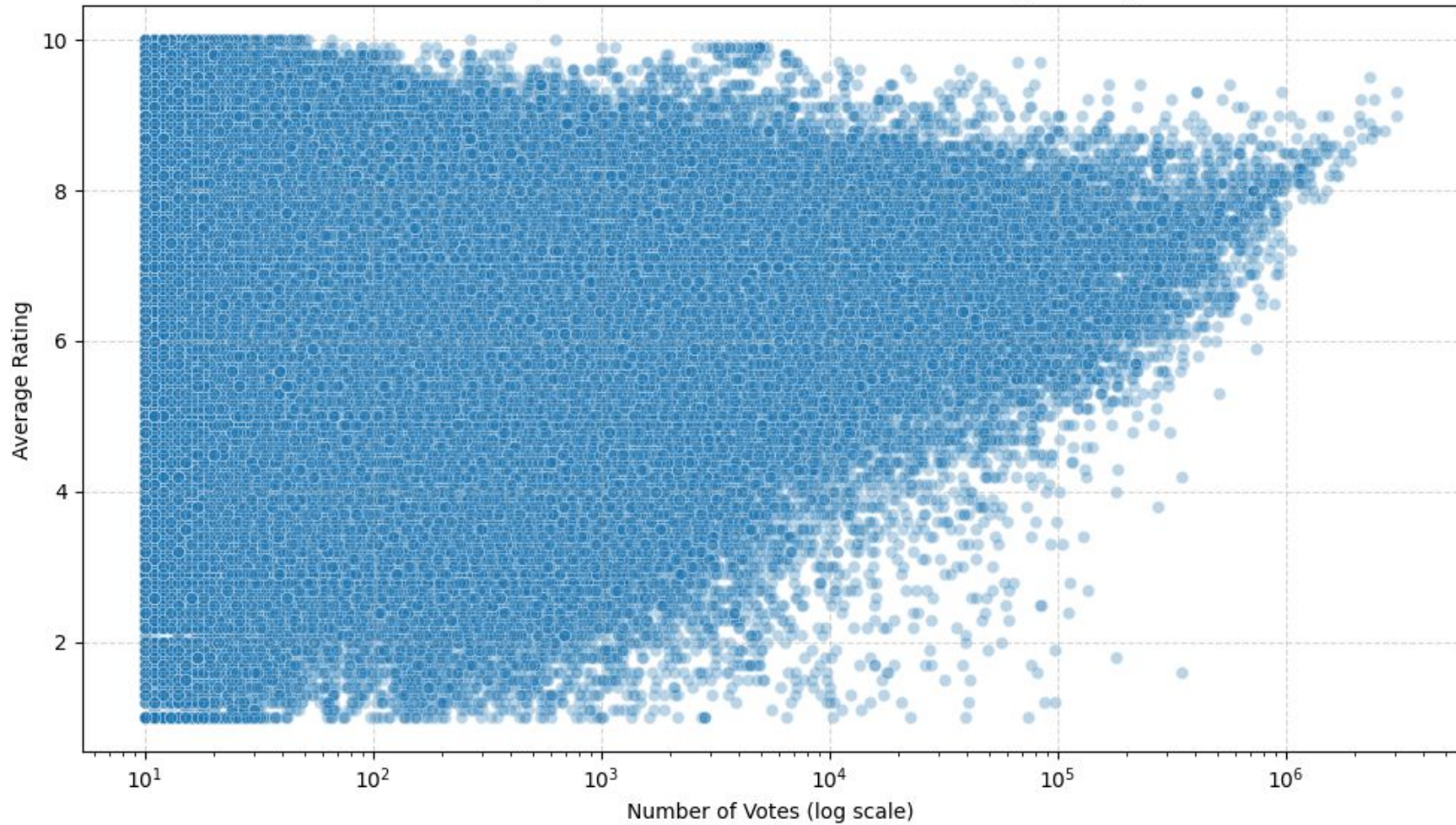
Average Rating & Number of Votes



Common Genres

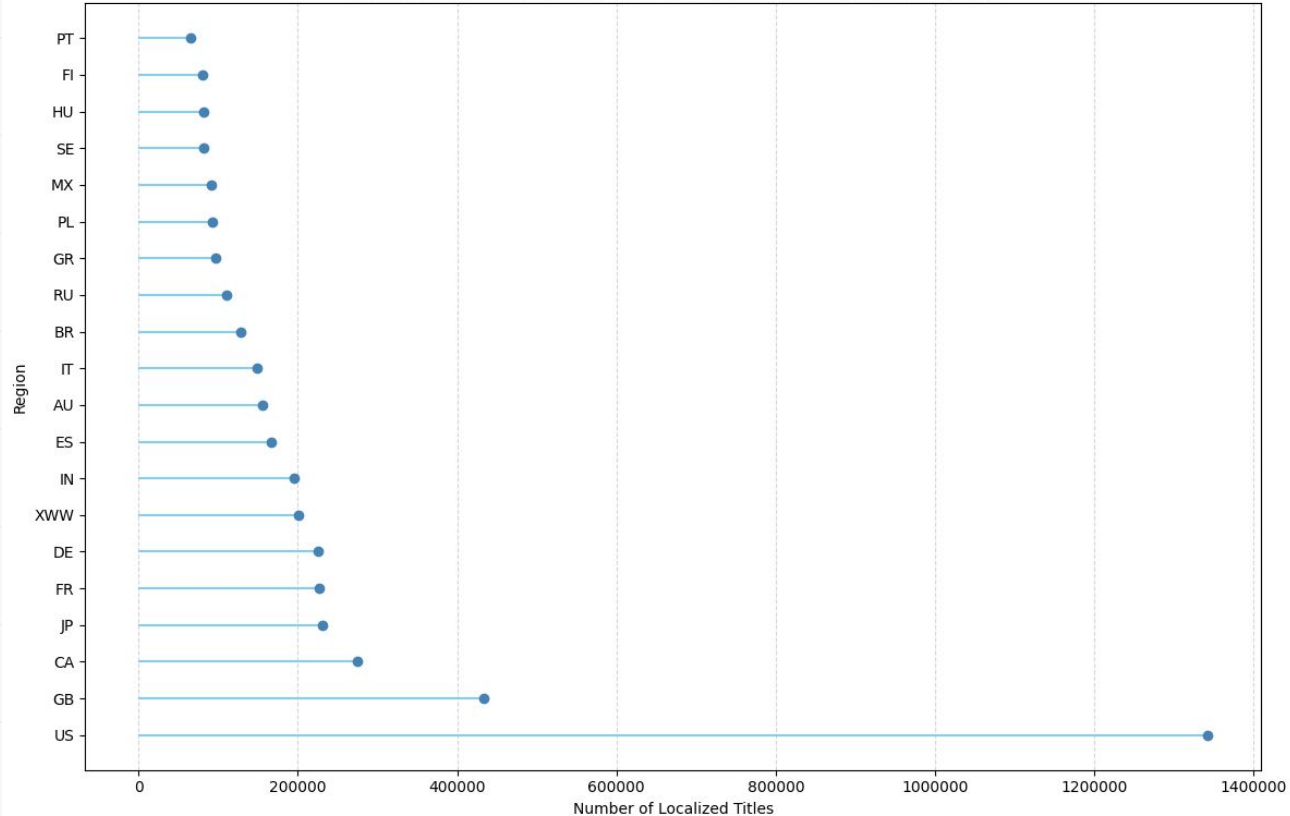


Relationship Between Number of Votes and Average Rating



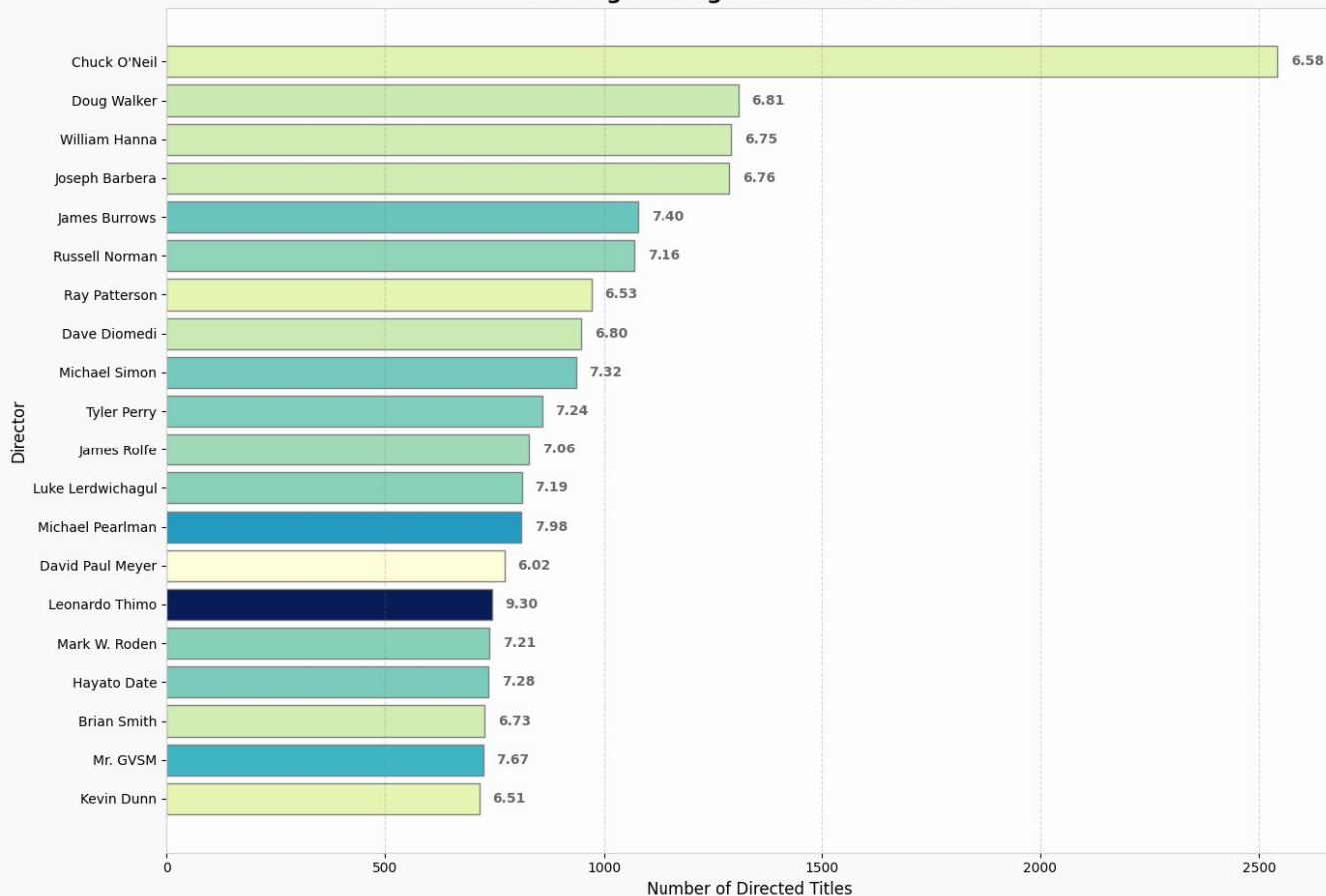
Popular Regions

Top 20 Regions by Number of Localized Titles (Lollipop Chart)

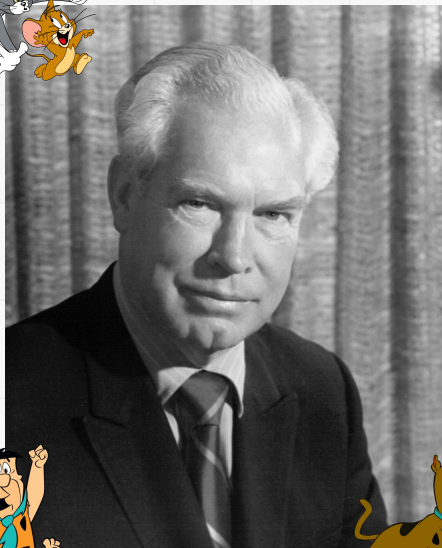


Prolific Directors

Most Prolific Directors (10+ Titles)
Average Rating shown next to bars



Director with the Largest Body of Work



William Hanna
titles 1294 ,
avg rating: 6.75



Doug Walker
titles 1310 ,
avg rating: 6.81

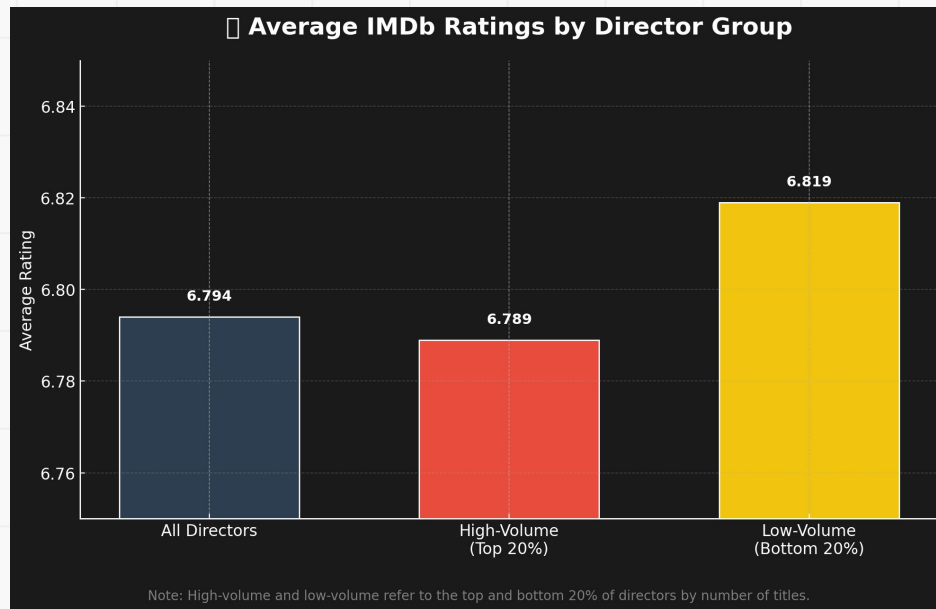


Chuck O'Neil
Titles 2543,
avg rating: 6.58



Directors Statistics

- Total directors analyzed: 317663
- Median projects per director: 1.00
- Standard deviation of average ratings: 1.361



Most Prolific Actors



Brahmanandam
titles 1135,
avg rating: 5.96



Jack Mercer:
titles 1687,
avg rating: 6.12

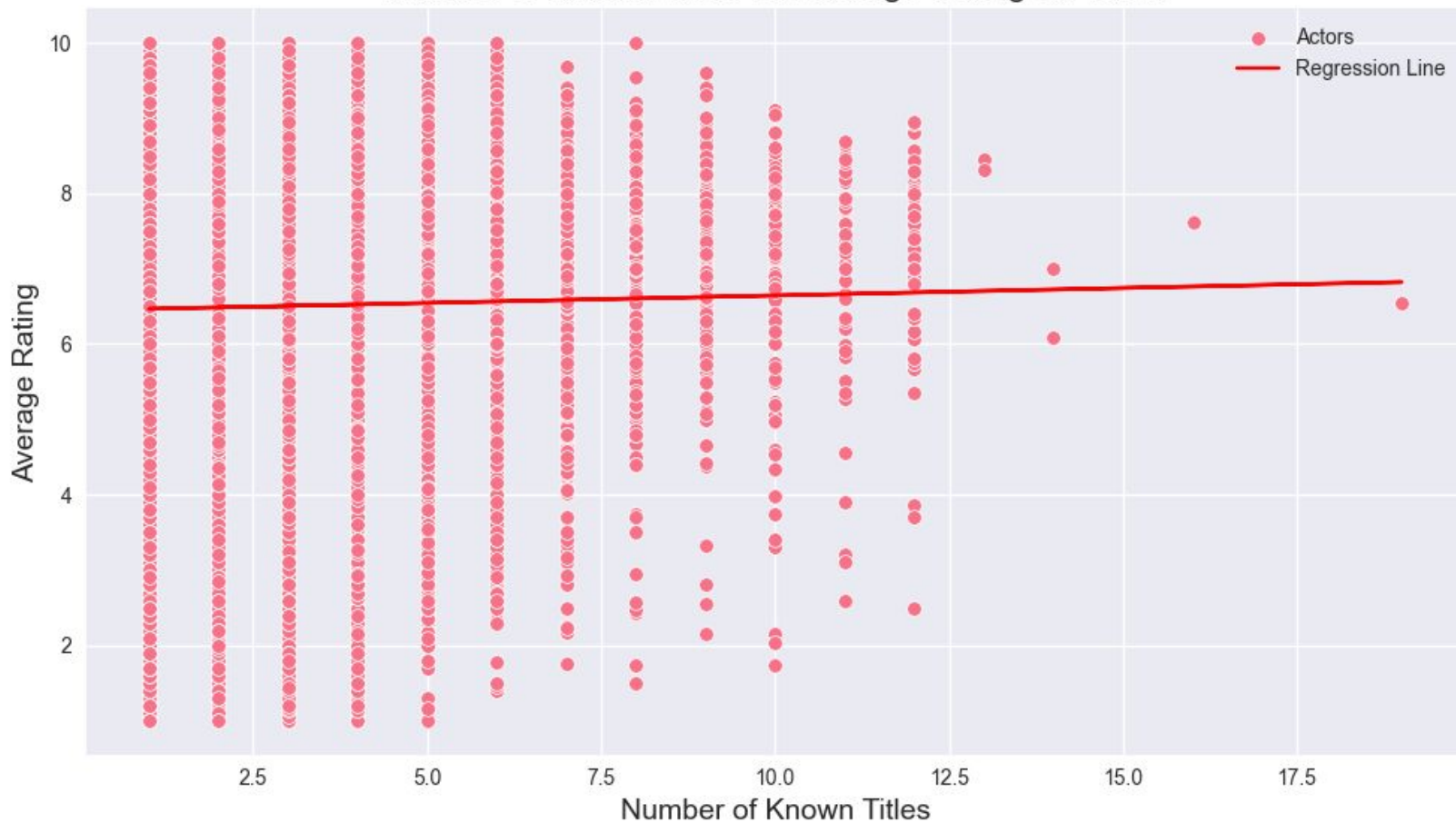


Mel Blanc:
titles 3701,
avg rating: 6.54





Number of Known Titles vs. Average Rating of Actors



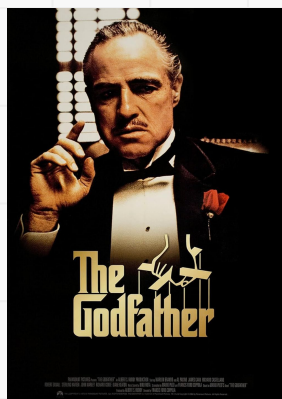


Top 10 Movies





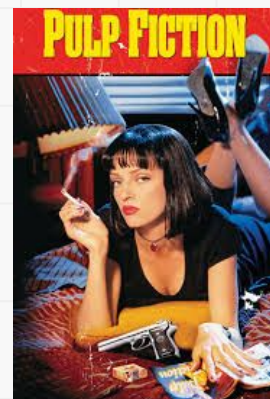
10 : place
Rate : 8.9
Votes : 2. 2 mill



9 : place
Rate : 9.2



8 : place
Rate : 8.7

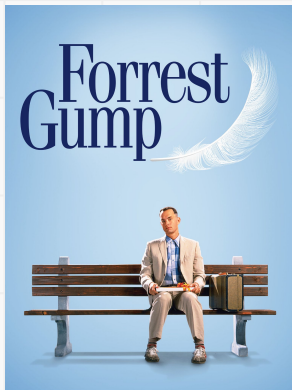


7 : place
Rate : 8.9



6 : place
Rate : 8.7

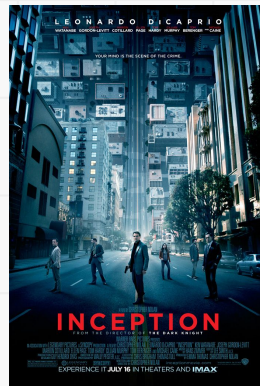




5 : place
Rate : 8.8



4 : place
Rate : 8.8



3 : place
Rate : 8.8
Votes: 2.5 mill



2 : place
Rate : 9.0
Votes : 3 mill



1 : place
Rate : 9.3
Votes : 3 mill





Top 10 TV Series





10 : place
Rate : 9.0
Votes : 0.75 mill



9 : place
Rate : 8.6



8 : place
Rate : 8.6



7 : place
Rate : 8.1

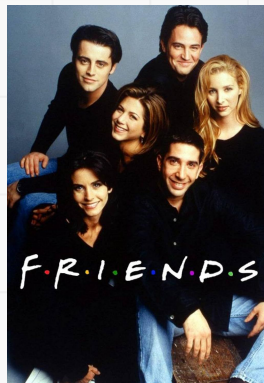


6 : place
Rate : 9.0
Votes : 1 mill





5 : place
Rate : 8.1



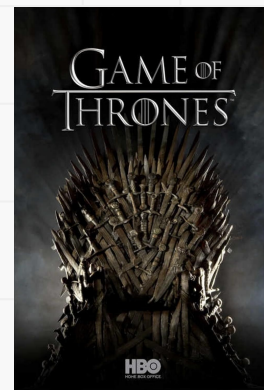
4 : place
Rate : 8.9



3 : place
Rate : 8.6



2 : place
Rate : 9.5
Votes : 2.3 mill



1 : place
Rate : 9.2
Votes : 2.4 mill



Conclusions

More doesn't always mean better

Actors and directors with more titles don't necessarily have higher average ratings.

High-rated content is diverse

Top-rated titles come from a wide range of genres and regions.

Surprising insights

- The most prolific directors often have lower average ratings.
- Some lesser-known actors appear in higher-rated titles.
- TV series frequently achieve higher ratings than movies.





Thanks!

Do you have any questions?